

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

Practical No: 14

Aim: Extracting date components using lubridate:: functions (R).

Code:

```
library(dplyr)
library(lubridate)
```

```
# --- 1. SETUP: Load and Simulate Dates ---
```

```
df <- read.csv("Cancer.dataset.csv") %>%
  select(id, diagnosis) %>%
  # Simulate a random diagnosis date for each patient within the year 2023
  mutate(
    Diagnosis_Date = as.Date("2023-01-01") + sample(0:364, n(), replace=TRUE)
  )
```

```
print("--- Data with Simulated Dates ---")
print(head(df, 3))
```

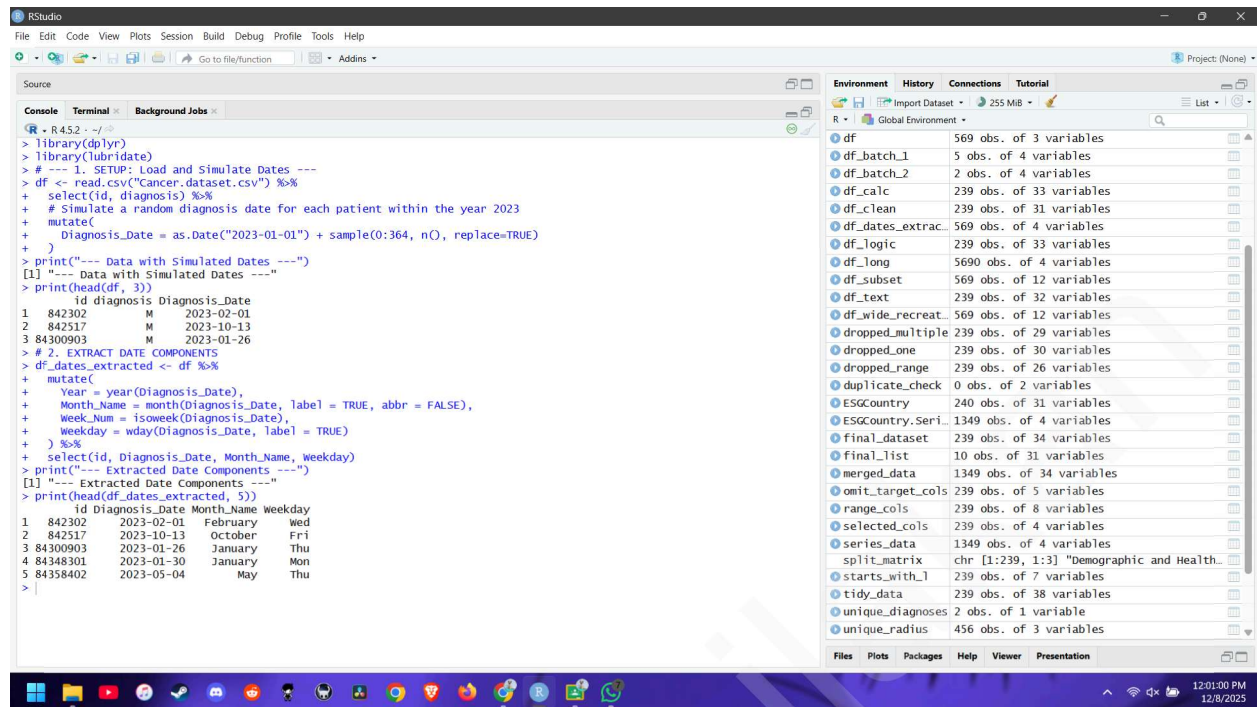
```
# 2. EXTRACT DATE COMPONENTS
```

```
df_dates_extracted <- df %>%
  mutate(
    Year = year(Diagnosis_Date),
    Month_Name = month(Diagnosis_Date, label = TRUE, abbr = FALSE),
    Week_Num = isoweek(Diagnosis_Date),
    Weekday = wday(Diagnosis_Date, label = TRUE)
  ) %>%
  select(id, Diagnosis_Date, Month_Name, Weekday)
```

```
print("--- Extracted Date Components ---")
print(head(df_dates_extracted, 5))
```

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Output:



The screenshot displays the RStudio interface. The console on the left shows the execution of R code that simulates data, extracts date components, and creates various data frames. The Environment pane on the right lists the objects created in the global environment, including data frames like 'df', 'df_batch_1', 'df_batch_2', 'df_calc', 'df_clean', 'df_dates_extracted', 'df_logic', 'df_long', 'df_subset', 'df_text', 'df_wide_recreat', 'dropped_multiple', 'dropped_one', 'dropped_range', 'duplicate_check', 'ESGCountry', 'ESGCountry.Seri', 'final_dataset', 'final_list', 'merged_data', 'omit_target_cols', 'range_cols', 'selected_cols', 'series_data', 'split_matrix', 'starts_with_1', 'tidy_data', 'unique_diagnoses', and 'unique_radius'.

```
R - R 4.5.2 - ~/R
> library(dplyr)
> library(lubridate)
> # --- 1. SETUP: Load and Simulate Dates ---
> df <- read.csv("Cancer.dataset.csv") %>%
+   select(id, diagnosis) %>%
+   # Simulate a random diagnosis date for each patient within the year 2023
+   mutate(
+     Diagnosis_Date = as.Date("2023-01-01") + sample(0:364, n(), replace=TRUE)
+   )
> print("--- Data with Simulated Dates ---")
[1] "--- Data with Simulated Dates ---"
> print(head(df, 3))
  id diagnosis Diagnosis_Date
1 842302      M      2023-02-01
2 842517      M      2023-10-13
3 84300903    M      2023-01-26
> # 2. EXTRACT DATE COMPONENTS
> df_dates_extracted <- df %>%
+   mutate(
+     Year = year(Diagnosis_Date),
+     Month_Name = month(Diagnosis_Date, label = TRUE, abbr = FALSE),
+     Week_Num = isoweek(Diagnosis_Date),
+     Weekday = wday(Diagnosis_Date, label = TRUE)
+   ) %>%
+   select(id, Diagnosis_Date, Month_Name, Weekday)
> print("--- Extracted Date Components ---")
[1] "--- Extracted Date Components ---"
> print(head(df_dates_extracted, 5))
  id Diagnosis_Date Month_Name Weekday
1 842302      2023-02-01    February    Wed
2 842517      2023-10-13    October     Fri
3 84300903      2023-01-26    January     Thu
4 84348301      2023-01-30    January     Mon
5 84358402      2023-05-04      May       Thu
>
```

Environment

Object	Size
df	569 obs. of 3 variables
df_batch_1	5 obs. of 4 variables
df_batch_2	2 obs. of 4 variables
df_calc	239 obs. of 33 variables
df_clean	239 obs. of 31 variables
df_dates_extracted	569 obs. of 4 variables
df_logic	239 obs. of 33 variables
df_long	5690 obs. of 4 variables
df_subset	569 obs. of 12 variables
df_text	239 obs. of 32 variables
df_wide_recreat	569 obs. of 12 variables
dropped_multiple	239 obs. of 29 variables
dropped_one	239 obs. of 30 variables
dropped_range	239 obs. of 26 variables
duplicate_check	0 obs. of 2 variables
ESGCountry	240 obs. of 31 variables
ESGCountry.Seri	1349 obs. of 4 variables
final_dataset	239 obs. of 34 variables
final_list	10 obs. of 31 variables
merged_data	1349 obs. of 34 variables
omit_target_cols	239 obs. of 5 variables
range_cols	239 obs. of 8 variables
selected_cols	239 obs. of 4 variables
series_data	1349 obs. of 4 variables
split_matrix	chr [1:239, 1:3] "Demographic and Health..."
starts_with_1	239 obs. of 7 variables
tidy_data	239 obs. of 38 variables
unique_diagnoses	2 obs. of 1 variable
unique_radius	456 obs. of 3 variables